

ROBERT FINNIE

SOFTWARE ENGINEER | C++ | C# | EMBEDDED LINUX | REAL-TIME SYSTEMS

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PROFESSIONAL PROFILE

Experienced Software Engineer with more than 20 years of software development across embedded and real-time systems, commercial products, defence and research software. Extensive development experience in C++, C# and C, including work at Cranfield University and Mastergrave Ltd spanning embedded Linux, hardware/software integration, system architecture and end-to-end product development. Experienced in designing, implementing, integrating, delivering and supporting software for operational and customer-facing environments. Combines analytical problem-solving with a practical understanding of hardware, software and end-user requirements. Additional experience in consultancy, technical leadership, training, commercial application development and business management.

CORE TECHNICAL SKILLS

Programming: C++, C#, C, Python, JavaScript, Pascal, x86 Assembly and scripting languages.

Embedded & real-time systems: Embedded Linux, embedded and real-time software, bare-metal development, firmware, hardware/software integration, PLC/Arduino control systems and motion control.

Software engineering: Software architecture, application design, implementation, debugging, integration, maintenance, technical problem-solving and software lifecycle support.

Platforms & systems: Linux, UNIX, PC/Windows environments, Android, embedded platforms, robotics, GIS and Command and Control (C2) systems.

Application development: Commercial software, multi-user systems, simulation, Unity, interactive 3D applications and AI-driven projects.

Technical delivery: Technical and design leadership, consultancy, customer-focused development, technical training, postgraduate supervision and cross-disciplinary collaboration.

RELEVANT SOFTWARE ENGINEERING EXPERIENCE

Product Design & Developer | Mastergrave Ltd

January 2011 – January 2017 • Bedford, Bedfordshire

- Designed and developed a complete CNC-based engraving system for a commercial client, spanning hardware integration and end-user software.
- Worked across embedded Linux development, low-level hardware interfacing, real-time control and application-level functionality.
- Developed PLC/Arduino-based control systems, firmware and motion-control logic, integrating software with physical hardware.
- Developed a consumer-facing GUI using Python and C, connecting low-level control functionality with an accessible end-user application.
- Resolved technical problems across hardware/software boundaries to deliver an integrated commercial product.

Software Developer / Research Software Engineer | Cranfield University – Microprocessor Applications Group

February 1994 – October 2011 • Cranfield, Bedfordshire

- Developed software for defence and research organisations, including the UK Ministry of Defence and NATO.
- Applied C++ and C# extensively, alongside C and other languages, across embedded and real-time systems, GIS, Command and Control (C2) applications and large multi-user systems.
- Contributed to system architecture, technical design and implementation, translating operational and technical requirements into software solutions.
- Worked across software development, integration, deployment and ongoing support for complex systems.
- Provided technical and design leadership on applications deployed in operational environments, including Bosnia (1995) and Afghanistan (2002).
- Designed and delivered university-level courses in C programming and parallel computing for MSc students; supervised MSc theses and served as a viva examiner.

Consultant – SPECTRA Operational & Training Software | Royal School of Signals (UK Ministry of Defence)

January 2011 – January 2015; ongoing consultancy/support • Blandford Forum, Dorset

- Provided consultancy, software development, maintenance and support for operational software used by the Royal School of Signals.
- Supported SPECTRA, a specialist software platform originally developed during my time at Cranfield University and subsequently used in training and field exercises.
- Delivered on-site training and post-exercise analysis support for military personnel.

ADDITIONAL SOFTWARE DEVELOPMENT EXPERIENCE

Commercial Software Developer | Engine Creative – Hillarys Blinds project

October 2023 – March 2024

- Led development of a commercial 3D interior design and product-visualisation application using Unity and C#.
- Delivered the application for standalone Android and web platforms, combining interactive 3D functionality with a responsive JavaScript/React interface.
- Worked with an AWS-powered content management system to manage product data, assets and configurations across platforms.

Independent Consultant & Developer | Self-employed

January 2017 – Present • Bedford, Bedfordshire

- Continued software development through independent projects and consulting, including VR applications and AI-driven games.
- Developed and self-published a VR flight simulator/combat game for PCVR and Oculus platforms.
- Created a retro-style football game using reinforcement learning and machine-learning techniques for adaptive opponent AI.
- Combined software development with independent technical problem-solving and project delivery.

ADDITIONAL PROFESSIONAL EXPERIENCE

Pub Landlord – Sabbatical & Small Business Management | The Three Cups

May 2017 – July 2022 • Bedford, Bedfordshire

- Managed day-to-day operations of a high-volume pub, including staff scheduling, customer service and supplier relationships.
- Improved operational efficiency through process automation, including stock-control or point-of-sale systems.
- Developed leadership, prioritisation, financial oversight, crisis-management and customer-engagement skills.

ADDITIONAL ACHIEVEMENTS

- Member of Team Stellar, winners of the MoD Grand Challenge 2008, developing UAV-based threat- and obstacle-detection systems.
- Published author/co-author on digital signal and image-processing topics.
- Enhanced Security Check (eSC), UK Ministry of Defence.

EDUCATION & PROFESSIONAL DEVELOPMENT

- Bedford College of Higher Education – BTEC HND in Mathematics and Computer Science.
- Cranfield University – Numerous professional accreditations.

Portfolio: [Finniebob.github.io](https://github.com/Finniebob)